

Tangible Interfaces zur Unterstützung bewusster Projektwechsel

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Abstract

(Frequent and sometimes unconscious task switching is a standard occurrence in computer-based, multitasking work, which often diverts attention negatively and reduces cognitive control. This thesis investigates whether physical tokens used in a tangible interface for tab switching between parallel digital tasks can make this more deliberate and improve perceived focus and control compared to a standard digital interface. A within-subjects study with (8?) participants measures task switches, perceived focus and control, times and so on in two different interfaces to test this hypothesis.)

Keywords

Tangible Interface

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1 Introduction

Modern digital work environments require individuals to continuously switch between multiple tasks, projects, windows, and tabs for example. Especially in knowledge work settings, task switching has become a central aspect of everyday computer interaction [16]. Users frequently alternate between different documents, communication tools, browsers, secondary tasks, and so on, often within very short time intervals and with changing information contexts. Previous research has shown that many of these switches occur automatically or unintentionally, leading to fragmented attention and reduced feelings of control and concentration for ongoing work processes [5] [2].

Studies on Multitasking and interruptions demonstrate that frequent task switching introduces cognitive costs and negatively affects focus and self-regulation [11]. Czerwinski et al. reported that knowledge workers perform a large number of task switches during everyday work, many of which are selfinitiated rather than externally forced [3]. Similarly, Mark et al. showed that interruptions and constant switching increase stress, frustration, and mental workload while reducing perceived control and sustained attention [12]. Other research further suggests that switching between tasks leaves residual attention on the previously performed activity, a

phenomenon referred to as "attention residue" [1][14].

Nowadays most used interfaces support task switching primarily through graphical click or touch interactions such as tabs, buttons, notifications and pop-up windows which allow rapid control actions. While mostly efficient, these interactions may also encourage impulsive and unconscious switching behavior because transitions can be executed almost instantly with minimal physical or cognitive effort.

Research topics like Tangible User Interfaces (TUIs) and Embodied Interaction proposes that incorporating physical interaction into digital systems may support more deliberate and conscious user behavior. Tangible interfaces integrate physical objects into digital interaction processes and therefore actively involve the body in interaction [7] [17]. Because of that previous work suggests that physical interaction can increase awareness, intentionality and user engagement compared to purely graphical interaction techniques [6] [4].

Until now, only limited research has investigated the potential influence on multitasking behavior and project switching through tangible interactions in digital knowledge work environments. While TUIs already have been explored in various different domains such as collaborative work, education and information visualization, their role in regulating task switching behavior remains underexplored.

The goal of this study is not to maximize productivity, but rather to address gaps and explore whether physical interaction can influence how individuals perceive and perform digital task switching. In particular, the work focuses on unintended task switches, perceived focus, and feelings of control during work on concurrent tasks. To investigate this, a tangible task-switching interface is compared to a traditional tab-click-based interface in an experimental digital multitasking scenario through a small within-subject study. This paper presents therefore a simple prototype for tangible interactions and explores how physical interaction may influence task-switching behavior, focus and perceived control in named environments.

2 Related Work

Czerwinski et al. already found that task switching is a central characteristic of modern knowledge work, where users alternate between applications, documents and communication tools. And that such switching occurs very frequently in everyday work while often being self-initiated rather than externally triggered which highlight the highly fragmented nature of digital work environments [3].

Prior research also shows that task switching comes with measurable cognitive costs because of switched tasks requiring reorientation of attention and reconstruction of the previous mental context,

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which reduces efficiency and increases workload [8]. The concept of "attention residue" further explains that part of an individual's attention remains on the previous task after switching, negatively affecting performance on the current task [1] [14].

Additionally, interruptions contribute to increased stress and reduced perceived control. Mark et al. show that frequent happenings like this increase frustration, workload and time pressure, even when they may appear beneficial for short-term productivity [12]. In modern digital environments notifications and pop-ups further reinforce repeated switching behavior [9] [10].

Through TUIs investigations were made for how physical objects can represent and control digital information. The concept of "Tangible Bits" introduced the idea of coupling digital data with physical artifacts to enable more embodied forms of interaction [7] [17]. Embodied interaction theory suggests that cognition is closely linked to physical action and environmental engagement. In this context, TUIs aim to make interaction more physically grounded compared to traditional screen-based interfaces [7].

Known is furthermore that tangible interaction can increase awareness and intentionality. Fitzmaurice et al. argue that graspable interfaces support more direct manipulation [4], while Hornecker and Buur highlight improved engagement and awareness through physical interaction [6] and Marshall further suggests that physical interaction can slow down actions and make user behavior more deliberate [13].

Of particular relevance is the work by Okundaye et al. (2017), which introduces a system that links digital work contexts to physical RFID-based artifacts. Their findings show that physical representations can support context switching and help users re-access and organize digital work contexts. Study participants described the physical cards as more direct and tangible compared to traditional graphical interaction and reported a more intentional and noticeable switching feeling. The authors further argue that physical artifacts may support spatial organization and peripheral awareness of ongoing work contexts in ways traditional tab- or window-based interfaces cannot easily provide. However, users also noted practical limitations such as the required physical space and scalability of physical artifacts [15].

3 Study Design

3.1 Study Type

The here conducted study follows a within-subject experimental design to investigate the effects of a tangible interface on task switching during computer-based, multitasking workload. Participants perform the same set of concurrent tasks under two conditions:

- **Tangible Interface:** Switching between tasks is performed through physical tokens that represent the different tasks through QR-Codes.
- **Digital Interface:** Switching between tasks is performed through a conventional graphical user interface by using tabs through mouse-clicking.

The order of conditions is counterbalanced to reduce learning effects. In both interfaces, participants receive identical tasks, visualizations, interruptions and information with the only difference

being the interaction method used to switch between tasks.

As results quantitative and qualitative data is collected which include interaction logs, task-switching frequency, different timestamps, questionnaire responses, and observation logs.

Potential threats to validity embrace a small sample size, learning effects and a possible novelty effect of the tangible interface. Appropriate measures are taken during study design and data collection to account for these factors.

3.2 Research questions

As determined working continuously on concurrent tasks leads to frequent switches throughout work settings, often initiated automatically or in response to interruptions. This may lead to fragmented attention and reduced feelings of control. Therefore tangible interaction has been proposed as a way to support more deliberate and conscious actions by incorporating physical movement into digital workflows.

Based on this background, the study investigates the following research questions:

- **Research question 1 (RQ1).** Does a tangible interface for task switching reduce unintended task switches during digital multitasking work settings compared to a traditional graphical user interface?
- **Research question 2 (RQ2).** Does a tangible interface increase users perceived focus and sense of control during task switching in digital multitasking work settings?

4 Hypotheses

Based on what has been described the following hypotheses are proposed:

Hypotheses related to RQ1:

- **Hypothesis 1 (H1):** Participants will perform fewer unintended task switches when using the tangible interface instead of the graphical user interface.
Rationale. The physical action required to switch tabs makes transitions more deliberate and may reduce impulsive reactions to interruptions and distractions.

Hypotheses related to RQ2:

- **Hypothesis 2 (H2):** Participants will report higher levels of focus and perceived control when using the tangible interface instead of the graphical user interface.
Rationale. Physical interaction increases awareness of task changes and makes the active task more explicit, potentially supporting attention management and self-regulation.

5 Method

5.1 System Implementation

A browser-based prototype was developed using JavaScript, HTML and CSS and executed on a laptop with a connected mouse and webcam. The system simulates a multitasking workspace consisting of three concurrent tasks, which must be fully executed, represented as tabs within the interface. This prototype contains two interaction conditions: Tangible Interface Condition and Digital Interface Condition.

In the tangible condition furthermore the tangible interface, participants switch between projects using physical tokens equipped with QR-Codes. A webcam positioned above the designated interaction area detects the QR-Codes and activates the corresponding project tab within the application. Webcam and interaction area are always located on the side of the participant's dominant hand (See Figure 1).

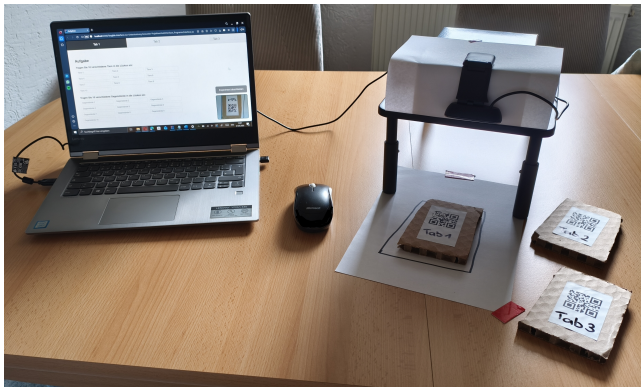


Figure 1: Experiment Setup.

Only one token can be active at a time, accordingly only one tab may be opened, and only the task located therein can be worked on. Moving a token into the active zone triggers a tab switch and automatically updates the interface. This interaction method is intended to make switching actions more deliberate and physically noticeable.

In the digital condition, which can be regarded as a widely adopted and known standard interface, participants switch between tasks by clicking on the corresponding tabs using a computer mouse. The visual layout, tasks, notifications, and displayed information remain almost identical to the tangible condition, the main difference being the interaction modality used for switching between tasks respectively tabs.

Furthermore, the system tracks and saves all relevant interactions during the experiment in a CSV file, including: task-switching events, timestamps, active task durations, reaction times, notification responses, task completion time and given answers. To simulate realistic multitasking environments, the prototype also outputs urgent pop-up notification and notification bubbles tied to tab 3. These interruptions are identical across both conditions.



Figure 2: Notifications used in the study interface.

5.2 Tasks And Experimental Materials

Participants work on three concurrent tasks designed to simulate different forms of cognitive workload and fragmented knowledge work.

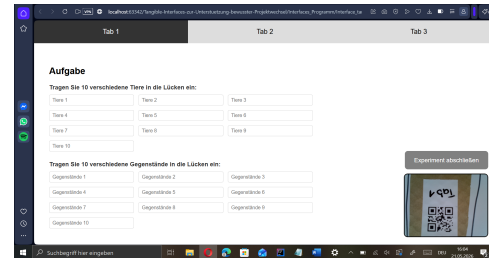


Figure 3: Example of Tab1 used in the study interface.

The primary main task (See Figure 3) is situated in tab 1 and requires sustained attention while participants are asked to complete two sets of fill-in tasks consisting of ten input fields each. The questions differ between the two interface conditions to reduce memorization and learning effects. This is intentionally designed to be simple and cognitively lightweight in order to keep main focus on task-switching behavior rather than task difficulty. Prompts include naming food items, colors, animals and objects.

While interruptions from the other tabs occur throughout the experiment execution, participant type their answers into the provided input fields which serves as a continuous baseline activity.

Tab 2 contains a small interaction based side quest asking for immediate attention triggered by the preset appearing of an notification pop-up. Participants can switch to the corresponding tab and complete short click-based actions or continue what they are working on. Here the project tries to create interruption pressure and opportunities for impulsive or unintended task switches which are commonly found in digital work settings.

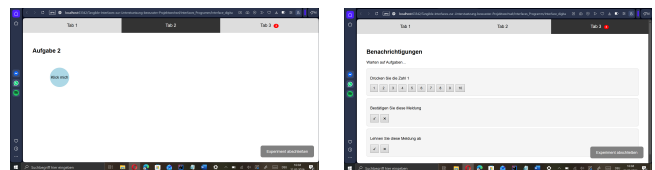


Figure 4: Examples of Tab2 and Tab3 used in the study interface.

For the last task, tab 3 is showing smaller, concise prompts that are completed by clicking the right answer. While participants work on other work tasks, small notification bubbles indicate that new content is available in the corresponding tab and can be solved immediately or ignored for the time. In contrast to the feigned urgency for tab 2, this intends to simulate background tasks that compete for attention without creating strong urgency.

The experimental environment for an interface condition is completed once all tasks have been executed or answered therefore

enabling the finish button which also triggers the download of a CSV file with the collected data.

5.3 Variables And Collected Data

The independent variable of the study is the interaction modality used for task switching:

- tangible task switching using physical tokens,
- digital task switching using click based tab interaction.

Therefore the dependent variables are several objective and subjective dependent variables which are collected during the experiment. Here the collected data consists of objective measures the interface prototypes record and collect in CSV files, like time and number of task switches, time spent on each tab/task, reaction time to notifications, inputs, switching sequences and timestamps. These measurements are used to analyze the switching behavior and attention management during multitasking.

Subjective measures are also taken by collecting short questionnaire answers using Likert-scale ratings and open questions about perceived focus and control, mental workload and awareness of switching behavior. Additionally, observation notes are documented during the study.

5.4 Procedure

Each experiment session follows the same procedure and lasts approximately 20 to 30 minutes.

- (1) Participants receive an overall introduction to the study and provide informed consent.
- (2) Participants receive a short introduction to the interfaces and task process. Instruction is given to use the system in a natural way that reflects how they would normally work.
- (3) A short pre-experiment questionnaire is answered by the participants.
- (4) Participants perform the multitasking scenario in the first assigned condition. After completion, they fill out the post-condition questionnaire.
- (5) The same scenario is then performed by the participants using the second interaction condition. A second post-condition questionnaire is completed.
- (6) Finally, a short concluding questionnaire is filled out by the participants comparing both interaction methods and giving demographic data.

The study targets around eight participants, four of whom already participated in a Pre-Study and mainly consists of young adults who are familiar with computer-based multitasking work environments. Participation is voluntary and all participants sign informed consent forms before the study begins.

5.5 Data Analysis

All the collected data from interaction logs and questionnaires are analyzed descriptively and comparatively between both conditions. Measures such as average switching frequency, triggered switches, reaction times, subjective ratings and so on are compared to identify behavioral tendencies and recurring perceptions.

5.6 Pre-Study

A small preliminary study was conducted with four participants prior to the main study. This pilot setup followed the same procedure as the main experiment and was used to test the overall experiment, tasks, interaction design, and questionnaires. Based on participant feedback and observations during this test run, the questionnaire was revised to improve clarity. In addition, minor technical issues in the software prototype were identified and fixed. Furthermore the conducted pilot study ran largely as intended and closely resembled the final study procedure.

6 Results

Results of statistical analyses and reports what was found(objectively)

7 Discussion

Interpretation of results, connection to research questions/hypotheses, with previous studies, limitations

8 Conclusion

Summary

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